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EVIDENCE FOR PHENOTYPE MATCHING IN SPINY MICE (*ACOMYS CAHIRINUS*)

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Abstract. Two series of experiments were conducted to assess the influence of phenotypic familiarity on the development of social interactions in spiny mice (*Acomys cahirinus*) weanlings. Animals that had a particular artificial odorant applied either to themselves alone, or to themselves and their littermates, subsequently displayed preferential responsiveness to unfamiliar animals that had been treated in the same manner with the identical odour. Likewise, littermates who were separated from each other for 9 days but each housed with another pup from their same litter subsequently responded preferentially towards one another. The phenotype of familiar littermates appears to be a significant factor in the development of social preferences.

Sibling recognition has been documented in a variety of invertebrate and vertebrate species, including sweat bees, *Lasioglossum zephyrum* (Kukuk et al. 1977; Greenberg 1979), social wasps, *Polistes metricus* (Ross & Gamboa 1981), tadpoles, *Rana cascadae*, *Bufo americanus* (Waldman & Adler 1979; Blaustein & O'Hara 1981), Canada geese, *Branta canadensis* (Radesater 1976), house mice, *Mus musculus* (Kareem & Barnard 1982), deermice, *Peromyscus leucopus* (Grau 1982), voles, *Microtus ochrogaster*, *M. pennsylvanicus* (Wilson 1982), spiny mice, *Acomys cahirinus* (Porter et al. 1978), ground squirrels, *Spermophilus* spp. (Holmes & Sherman 1982; Davis, in press), and pigtail macaques, *Macaca nemestrina* (Wu et al. 1980). In theory, the ability to recognize one's siblings could arise through exposure and familiarity or be genetically encoded on recognition alleles (e.g. Dawkins 1976; Bekoff 1981; Holmes & Sherman 1982). As evidence for the latter recognition mechanism, Yamazaki et al. (1976, 1978) have shown that house mice can recognize conspecifics of their own inbred strain through genetically-dependent odour cues. To date, however, systematic laboratory studies of the development of kin interactions in spiny mice offer no support for the hypothesis that sibling recognition in this species is mediated by recognition alleles (Porter et al. 1978, 1981; Porter & Wyrick 1979). Rather, weanlings respond preferentially to familiar conspecifics with whom they are raised — whether or not they are biological kin.

As recently pointed out by several authors (e.g. Dawkins 1976; Barash et al. 1978; Greenberg 1979; Bateson 1980; Bekoff 1981; Waldman 1981; Holmes & Sherman 1982), shared or familiar phenotypic traits could also allow for preferential social interactions among kin that have previously not been exposed directly to one another. For instance, an animal might respond selectively to conspecifics with whom it shared a specific visual or olfactory characteristic, or to those whose phenotype matched that of familiar individuals such as its mother or littermates. Thus *A. cahirinus* pups that suckle from the same female show a tendency to respond preferentially to one another, suggesting that lactating females may label their offspring with characteristic chemical cues (Porter et al. 1981). The present series of experiments investigates further the possible role of phenotype matching in the development of social preferences in spiny mice. The original impetus for studying the development of kin recognition in *A. cahirinus* was that it is unique among murid rodents in having precocial offspring. Neonatal spiny mice, which are capable of independent locomotion within hours of birth and whose sensory modalities are all functional at this age, are not confined to a nest during the pre-weaning period (e.g. Porter & Ruttle 1975). One might therefore predict that this species would have evolved a more rapidly developing kin recognition capability than is found in altricial rodents.

I. Odour Phenotype Matching

Matching of Odours Applied to the Test Animals and their Littermates

In comparison to control animals, laboratory rodents raised with artificially odorized dams (or with odorized littermates as well as dams) subsequently display heightened social responsiveness to conspecifics scented with the familiar maternal odour (Mainardi et al. 1965; Marr & Gardner 1965; Marr & Lilliston 1969; Carter & Marr 1970; Brunjes & Alberts 1979). Artificial odours were also used in the present study of odour phenotype matching. However, rather than odorizing the mothers of the subject animals, all pups in intact litters were scented following weaning and separation from their mother. The immediate specific question being addressed by this first experiment is whether weanlings would display preferential responsiveness to unfamiliar age-mates bearing the same olfactory cues as themselves and their littermates.

Subjects. Subject litters of three to four pups were randomly selected from the laboratory breeding colony of *Acomys cahirinus*. Breeding pairs in the colony were housed in individual cages and the subject offspring remained with their parents until the beginning of the experimental treatment at weaning.

Odour exposure. All animals tested in experiment 1 were weaned at 16–21 days of age. At that time, three littermates were removed from their parents and placed together into a plastic cage measuring $18 \times 29 \times 13$ cm high (litters of four were culled to three pups at weaning). Each group of three littermates remained intact until the start of the test session.

Odours were applied by rubbing a saturated cotton swab on the back of each animal near the base of the tail. In this manner, all animals in a litter were odorized for the first time immediately after weaning (day 1 of treatment) and again on days 3, 5, and 6 of the treatment period. Testing began following the final odour application. A total of four artificial odorants were used: musk oil; oil of clove; lemon/lime; and cherry. Only one of these odours was applied to the three animals in any given litter.

Testing. Animals were tested in a glass terrarium measuring $29 \times 60 \times 30$ cm high lined with bedding material (sugar-cane waste) and containing mouse chow and fresh apple chunks. The 3-day test session began when four unfamiliar animals (i.e. one animal from each of four different litters) were placed together into an observation terrarium. In each instance, two of these animals (along with their littermates) had been treated with the same odorant, while the remaining two animals had been treated with a second odour during the treatment period. For example, two animals previously treated with clove might be tested along with two cherry-treated weanlings. The four animals comprising each test group had not been exposed to one another prior to testing and were born within 48 h of each other. To enable the observer to recognize individuals, all animals were coded by small colour marks sprayed on their backs. This marking technique has been used in a number of studies with *A. cahirinus* with no noticeable influence on the animals' social interactions (e.g. Porter et al. 1978, 1981; Porter & Wyrick 1979).

Overall, 18 groups of four animals each were tested. In nine of these groups, two unfamiliar clove-treated animals were housed with two cherry-treated animals during testing, while the remaining nine groups each contained two musk- plus two lemon/lime-treated weanlings.

Each group of animals was observed during six 5-min sessions per day over a 3-day period. The first observation session began 60 min after the animals had been placed into the observation cage to allow time for them to explore and become somewhat familiar with their new environment. At the beginning of each 5-min session and at the end of each 60-s interval therein, we recorded the identities of any animals that were physically touching one another (i.e. touching of head, limbs or trunk; excluding touching of whiskers or tails). Thus, for each 5-min observation session there was a total of six observation points. The six 5-min observation sessions were distributed throughout the day with at least 30 min between successive sessions. Over the 3 days of observation there was a total of 108 (36×3 days) sample points for each group.

As for previous studies of sibling recognition in *A. cahirinus*, dyadic touching was selected as an operational measure of social preferences since it is a relatively unambiguous reflection of mutual attraction (e.g. Porter et al. 1978, 1981). In contrast, when three or more animals are huddled together it is not possible to ascertain whether there is mutual attraction amongst all animals or whether A is attracted only to B, B to C, etc. Therefore, the data reported below pertain to social pairs; that is, two animals in physical contact with each other only.

Results. As delineated above, the relative frequency of dyadic touching by particular animals was adopted as an unambiguous index of social preferences. Table I contains a summary of the median frequencies of dyadic pairing by animals that had been treated with the same odour and by animals treated with two different odours (different-odour pairings). Since different-

Table I. Median Frequencies of Pairings over the 3-day Observation Period by Unfamiliar Animals Treated with the Same or Different Odours; Prior to Testing, All Animals had been Odorized in Groups Each Containing Three Littermates

	Median frequencies of pairings	
Odour combinations during testing	Animals treated with the same odour	Animals treated with different odours
Clove/cherry ($N = 9$ groups)	71 (Mean = 64.6) (Range = 0–161)	0 (Mean = 8.8) (Range = 0–46)
Musk/lemon–lime ($N = 9$ groups)	26 (Mean = 29.6) (Range = 0–92)	1 (Mean = 1.6) (Range = 0–6)
Total ($N = 18$ groups)	39 (Range = 0–161)	0.5 (Range = 0–46)
Overall median, excluding groups with no dyadic pairings ($N = 16$ groups)	42	1
Corrected overall medians ($N = 16$ groups)	63 (Range = 0–241.5)	0.75 (Range = 0–34.5)
	Wilcoxon's $T = 3.5$, $N = 16$, $P < 0.01$ (one-tailed test)	

odour pairings could arise from twice as many of the possible combinations of two animals (in each observation cage) as could same-odour pairings, the expected ratio of different- to same-odour pairings was 2:1 within each observation group. To correct for this expected 2:1 ratio, the observed frequency of different-odour pairings for each group was multiplied by 0.75 and the frequency of same-odour pairing by 1.5. The resultant corrected mean frequencies of same- versus different-odour pairings were compared with Wilcoxon's matched-pairs signed-ranks test. The units for this analysis were the frequencies of dyadic pairings for each group of four animals rather than the scores for the individual subjects. Two of the 18 groups had no occurrences of dyadic pairings (neither by same- nor different-odour animals). Therefore, although a total of 18 groups were tested in experiment 1, the N for the statistical analysis = 16, as tied scores are deleted from the Wilcoxon's text.

It is evident from an inspection of Table I that unfamiliar weanlings that had been treated with the same odour (applied to themselves and their littermates) during the treatment period subsequently paired significantly more often than did different-odour animals. A similar pattern of results was obtained for both the clove–cherry and the musk–lemon/lime odorant groups. Therefore, the data for the two odour combinations were combined for the statistical analysis.

Matching of Odours Applied to Isolated Individual Animals

In the previous experiment, training odours were applied directly to all three littermates housed together during the treatment period. Therefore, the observed preferential pairings by unfamiliar same-odour animals could have been mediated through matching of the individual subjects' own (artificial) odour, matching of the familiar littermates' odour, or a combination of these two processes. Experiment 2 was conducted to determine whether weanlings would show evidence of preferential responsiveness to age-mates whose odour matched that applied to themselves only (i.e. odours were applied to individually isolated animals rather than group-housed litters).

Methods. Subjects were randomly selected from amongst litters of three to four pups born in the laboratory breeding colony of *A. cahirinus*. Between days 16 and 21 postpartum, pups were weaned and placed individually into isolation cages (one pup per cage) measuring 18 × 29 × 13 cm high, where they remained throughout the treatment period. A total of 72 pups were tested, with 18 assigned to each of the four odorant conditions, i.e. musk, clove, lemon/lime and cherry. The appropriate odour was applied to each pup for the first time on the day of weaning (treatment day 1) and again on days 3, 5, and 6 of the treatment period. Testing began immediately following the final odour application (on treatment day 6).

As for experiment 1, four unfamiliar animals (not previously exposed to one another) were placed together into an observation terrarium at the beginning of the 3-day session. Nine of these four-animal groups contained two musk- and two lemon/lime-treated weanlings, while the remaining nine groups each contained two pups to which clove had been applied during the treatment phase, along with two cherry-treated pups. Thus, in each group of four animals ($N = 18$ groups), two animals had previously been treated with an identical odour and the other two had been treated with a second odour that differed from that applied to the former animals but was the same for these latter two weanlings. Further procedural details regarding treatment and testing were otherwise identical to those discussed above for experiment 1.

Results. The median frequencies of different- and same-odour pairings are presented separately for the clove/cherry and musk–lemon/lime odour groups in Table II. Since the results were similar for the two odour combinations, the data were combined for further statistical analysis. Following correction for the expected 2:1 ratio of different-versus same-odour pairings (as discussed for experiment 1), the corrected values for the nine groups with non-zero pairing frequencies were compared statistically using Wilcoxon's test. As seen in Table II, the frequency of pairings by animals bearing the same odour was significantly greater than that of animals each treated with a different odour.

II. Littermate Phenotype Matching

In the above experiments, odour phenotypes were manipulated by applying artificial odorants to spiny mouse weanlings. A final experiment was designed to investigate matching of naturally occurring phenotypic traits as a possible mediating mechanism for kin recognition.

Previous related research with spiny mice has shown that exposure and familiarity facilitate sibling recognition. Siblings that are housed together prior to testing subsequently display behavioural indications of recognizing one another (Porter et al. 1978). Although siblings that are individually isolated for up to 6 days subsequently recognize one another, such recognition is not evident following 8 days of isolation rearing (Porter & Wyrick 1979). Furthermore, if weanlings are separated from their littermates on day 20 and housed with unfamiliar unrelated age-mates for 10 days, such foster littermates respond to each other preferentially while biological siblings display no signs of recognition (Porter et al. 1981).

The specific question addressed by the present experiments is whether recognition by separated littermates would occur if the littermates in

Table II. Median Frequencies of Pairings over the 3-day Observation Period by Unfamiliar Animals Treated with the Same or Different Odours; Prior to Testing, Animals had been Odorized in Individual Isolation Cages

Odour combinations during testing	Median frequencies of pairings: Animals treated with the same odour	Median frequencies of pairings: Animals treated with different odours
Clove/cherry ($N = 9$ groups)	6 (Mean = 9.4) (Range = 0–33)	0 (Mean = 0.2) (Range = 0–2)
Musk/lemon–lime ($N = 9$ groups)	0 (Mean = 18.1) (Range = 0–107)	0 (Mean = 0.0) (Range = 0)
Total ($N = 18$ groups)	0 (Range = 0–107)	0 (Range = 0–2)
Overall median, excluding groups with no pairings ($N = 9$ groups)	17	0
Corrected overall medians ($N = 9$ groups)	25.5 (Range = 0–160.5)	0 (Range = 0–1.5)
	Wilcoxon's $T = 1.5$, $N = 9$, $P < 0.01$ (one-tailed test)	

question were each housed with another pup from their litter during the separation period. That is, whether phenotypic cues of one's littermates and self can function as standards to mediate the recognition of unfamiliar siblings who presumably share such traits.

Matching of Phenotypic Cues Associated with Individual Test Animals and Their Littermates

Subjects. Subject litters containing 4–5 pups were selected on a random basis from the laboratory breeding colony of *A. cahirinus*.

Littermate exposure. Between days 18 and 20 after birth, four pups from each of two litters (i.e. a total of eight pups) were removed from their respective home cages and housed two littermates per cage. Thus, two of these cages each contained two pups from one litter, while the remaining two cages each contained two pups from a second litter. All pups were marked for purposes of individual identification and food and water were available ad libitum. The paired animals remained together for nine full days, with testing beginning immediately after this exposure period.

Testing. At the beginning of the test session, four animals — one from each of the above four cages — were placed together into a single observation terrarium. The remaining four animals (again, one from each exposure pair) were placed together into a second terrarium. An observation terrarium therefore contained two pairs of littermates (i.e. four pups from two different litters). In each instance, the pups making up a littermate pair had been housed apart for the 9 days immediately preceding the start of testing. During that same 9-day period, however, each weanling had been housed with another of its littermates. Thus, littermates housed together during the exposure period were not together during testing.

Overall, 10 groups of four animals each were observed over 5-day test periods. Further procedural details are otherwise identical to those described above for the previous experiments.

Results. Dyadic pairings were observed in all but one of the 10 groups of animals. The frequencies of pairings by biological siblings (littermates) and non-siblings are summarized in Table III. Since non-sibling pairings were twice as likely as sibling pairings by chance alone (i.e. in each observation cage there were only two possible sibling pairs but four possible combinations of animals in non-sibling dyads), the frequencies of both classes of pairings were corrected as described above prior to the statistical analysis. Comparison of the corrected frequencies indicates that sibling dyads were observed significantly more often than were non-sibling dyads, which we take as evidence for sibling recognition.

Discussion

The data presented above suggest that familiar social odours can influence interactions among *A. cahirinus* weanlings by serving as a learned

Table III. Median Frequencies of Pairings by Non-siblings and 'Unfamiliar' Siblings Following a 10-day Training Period During which Each Test Pup was Paired with a Littermate Not Included in the Subsequent Test Session

	Median frequencies of pairings		
	Sibling dyads*		Non-sibling dyads†
Overall median ($N = 10$ groups)	6 (Range = 0–143)		0 (Range = 0–13)
Median excluding one group with no pairings ($N = 9$ groups)	6 (Range = 1–143)		0 (Range = 0–13)
Corrected median ($N = 9$ groups)	9 (Range = 1.5–214.5)	Wilcoxon's $T = 3$, $N = 9$, $P < 0.01$ (one-tailed test)	0 (Range = 0–9.75)

*Littermates housed apart during training.
†Unrelated animals not previously exposed to one another.

standard against which odours of conspecifics are compared. In the first series of experiments, weanlings responded preferentially to unfamiliar age-mates whose odour was artificially similar both to their own and that of their littermates with whom they had been housed prior to testing. Furthermore, unfamiliar weanlings that shared an odour only with one another and not with their respective littermates as well likewise displayed selective interactions. However, when, as in the final experiment, the salience of naturally occurring cues for phenotypic matching is assessed, the pattern of results differs from that obtained when phenotypic similarity is manipulated with artificial odorants. Littermates that were housed apart but reared with another pup from their same litter subsequently recognized one another. This is in marked contrast with previously reported data indicating that pups isolated for 8 days prior to testing show no reliable littermate preferences (Porter & Wyrick 1979). Preferential responsiveness to 'unfamiliar' siblings following exposure to littermates is not a function of social contact per se during the pretesting period: previous studies (Porter et al. 1981) have found that cross-fostered weanlings respond preferentially to their foster littermates, while displaying no evidence of recognizing their original siblings from whom they were separated at the time of fostering. Thus, whereas spiny mouse weanlings appear to recognize familiar age-mates, or those presumably sharing phenotypic traits with familiar age-mates, prolonged isolation obviates such selective social responsiveness.

As seen above, although animals that were scented with the same odorant while in isolation subsequently responded preferentially to one another, untreated littermates isolated for a comparable period of time display no evidence of recognition (Porter & Wyrick 1979). The difference in behaviour between these two groups of isolates could be due to differential similarity of the recognition cues in these two experiments. The artificial odours may have been more intense or distinctive than the natural odours available in the latter experiment. In addition, while animals scented with the same artificial odorant thereby shared an identical recognition cue, the natural odours of littermates (or cues in other sensory modalities) are likely to be similar but not identical. Identical odours might be more readily matched or recognized than the less similar cues. In any event, it appears that the phenotype of familiar littermates may be a significant factor in the development of social preferences.

Results similar to the above have recently been reported for sweat bee nest guards, which selectively admit conspecifics whose odour matches that of the guards' own nestmates (Buckle & Greenberg 1981). However, there is no evidence that the nest guards use their own odours as standards for nestmate recognition, which is consistent with our findings with naturally-occurring phenotypic cues. Ground squirrels (*Spermophilus parryii*) likewise appear to use littermate characteristics to facilitate recognition of kin (Holmes & Sherman 1982). In laboratory tests, aggression among unfamiliar ground squirrel sisters was negatively correlated with the number of siblings with whom those sisters had been raised.

In theory, familiar phenotypic cues serving as standards to mediate kin recognition need not be rigidly genetically determined: they could vary according to environmental influences and still function as effective standards. However, for the evolution of kin recognition through phenotype matching, it is necessary that phenotypic similarity correlate positively with genetic relatedness (e.g. Barash et al. 1978). In spiny mice, maternal labelling of offspring is an example of (presumably) environmentally labile chemical cues serving to mediate kin recognition. Previous research (Porter et al. 1981) has shown that pups that suckle from the same female subsequently prefer to interact with each other even if they have not previously been directly exposed to one another. While interacting with suckling pups, the lactating female apparently transfers odour cues, through her milk, saliva or urine, onto the body surface of the pups. These familiar maternal odour labels in turn enable the pups to recognize one another. As for maternal pheromone (Porter & Doane 1977), the characteristics of maternal labels are likely to vary with the diet of the female and therefore be somewhat malleable. Nonetheless, to the extent that females allow only their own offspring, or offspring of close relatives, to suckle, pups carrying the same maternal label would be siblings (or other classes of close genetic kin), and so phenotypic similarity would indeed be correlated with genetic relatedness. In the typical litter situation, of course, the influence of maternal labelling on the development of sibling recognition would be augmented by the effects of direct exposure to littermates. Thus, as pointed out by Holmes & Sherman (1982), kin recognition could develop from the interaction of several proximal mechanisms.

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